INCH-POUND

MIL-F-48425A (AR)
AMENDMENT 4
23 February 2001
SUPERSEDING
AMENDMENT 3
31 July 1989

MILITARY SPECIFICATION

FIN ASSEMBLY, M24

MIL-F-48425A (AR), was inactivated after 14 February 1997 for new design.

This amendment forms a part of MIL-F-48425A (AR), dated11 September 1980, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.2: Delete: "ASTM E-18, Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials."

* 2.2: Between ASTM Method E-18 and ASTM Method E-340 add the following:

"ASTM B-117 Standard Test Method of Salt Spray (fog) Testing"

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3.4: Delete: "fin housing" where it appear (2 places), and substitute "cartridge housing."

AMSC N/A FSC 1310

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4.3.1: Add the following to the list of components submitted for first article testing:

"Part Description	Drawings	Quantity
Flat tensile specimen	See 4.5.3.2	2
Round tensile specimen	See 4.5.3.2	2
Flat fully prepared micro-structure specimen	See 4.5.3.3.1	2
Round fully prepared micro-structure specimen	See 4.5.3.3.2	2
Macroetch specimens from 1 extruded bar of housing stock	See 4.5.2	2
Macroetch specimens from 1 extruded bar of fin stock	See 4.5.2	2
Blade security specimens	See 4.5.6	5"

Last 2 paragraphs: Delete in their entirety and substitute, "Macroetch specimens shall be identified as to their origin (front or rear of billet)."

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Table I, First Article Inspection:

Delete: "Hydrotest, to destruction, 10 (a), 3.2, 4.5.4."

Delete: "Extrusion Blanks (housing)" and substitute, "Fully heat-treated cartridge housing stock."

Delete: "Hardness test 2(c), 3.1, 4.5.3.1."

Microstructure examination, Delete: "1, 3.3, 4.5.3.3," and substitute, "2(C), 3.3, 4.5.3.3.2."

Delete: "Extrusion Blanks (Fin)" and substitute, "Fully heat-treated fin stock."

Add: "Microstructure examination, 2, 3.3, 4.5.3.3.1."

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Note a: Delete in its entirety.

Note c: Delete in its entirety and substitute, "(C) Microstructure specimens shall be taken from either end of each round tensile specimen submitted for testing."

Note d: Delete in its entirety and substitute "(d) Largest standard flat-type specimens available."

* Table I. In "Fin" section, under "Examination for Defects…" add:

"Salt spray test	2 e/	3.2	4.5.6"

* In "Housing" section, under "Examination for Defects..." add:

"Salt spray test	2 e/	3.2	4.5.6"

* Add note: "(e) Specimens for this test may be taken from the samples used for the above examination, or separate components may be used."

4.4.1.1: Delete in its entirety and substitute the following:

"4.4.1.1 Cartridge housing lot. A cartridge housing lot shall consist of blanks from the same heat treatment lot of the same alloy that have been heat treated in the same heat treating equipment at the same temperature in one unchanged process. A heat treatment lot may consist of more than one solution heat treatment batch; however, the samples for tensile and microstructure tests hereinafter prescribed must be selected from and represent each solution heat treatment batch. A solution heat treatment batch shall consist of one lot of the same alloy that has been thermally treated by the same method in the same furnace equipment at the same time and temperature in one unchanged process. The identification and segregation of each solution heat treatment batch is not required after acceptable compliance with the tensile, and microstructure test."

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4.4.1.2: Delete in its entirety and substitute the following::

"4.4.1.2 Fin stock lot. The extrusion lot shall consist of extruded stock of the same alloy that has been thermally treated by the same method in the same furnace equipment at the same temperature and time in one unchanged process regardless of the solution heat treating method utilized. Each extrusion lot representing material that has been solution heat treated after extrusion must be solution heat treated at the same time in the same furnace using the same temperature and time in the furnace. The extrusion lot thus constituted shall be the basis for selecting the sample blanks for the tension and microstructure tests hereinafter prescribed."

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4.4.2.b, Delete the following: "For inspection paragraph 4.4.2.4 Minor 205 (Length on rear thread, max.): Inspect 1 housing at the end of each day's production by cutting the rear of the housing in half (longitudinally). Length shall be measured from the rear face to the end of the thread's run out. Any housing which does not meet the drawing requirement shall cause the rejection of all material produced since the last successful inspection."

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4.4.2.1: Delete: "Fin and housing extrusion bars" and substitute, "Fin and housing extrusion bars prior to the final heat treatment."

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4.4.2.2: Delete: "Fin and housing extrusion blanks" and substitute, "Fully heat-treated fin and housing stock."

Special Category, Delete: "Special a, Hardness Test, 80, 3.1, 4.5.3.1"

Note a: Delete in its entirety and substitute the following:

"(a) Four specimens total (2 round from housing stock, 2 flat from fin stock). Failure of any sample shall cause rejection of the applicable solution heat treatment batch or press-quench lot (see para. 4.4.1)."

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Note b: Delete in its entirety and substitute the following:

"(b) Four specimens total (2 round, 2 flat). Failure of any sample shall cause rejection of the applicable solution heat treatment batch or press-quench lot (see para. 4.4.1)."

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4.4.2.4, Major 116: Delete "4.5.5.2" and substitute "Visual" in Inspection Method column.

Add Majors:

	"119 True position of cartridge cavity with pitch diameter of rear thread min.length	.40%	3.2	Gage
	120 Pitch diameter of forward thread, max.	.40%	3.2	Gage
	121 Pitch diameter of forward thread, min.	.40%	3.2	Gage
	122 Pitch diameter of rear thread, max.	.40%	3.2	Gage
	123 Pitch diameter of rear thread, min.	.40%	3.2	Gage
	124 Minor diameter of rear thread, max.	.40%	3.2	Gage
	125 Diameter of cartridge cavity.	.40%	3.2	Gage
*	126 Salt Spray Test	a/	3.2	Visual"

Minor 205: Delete "Tool control/4.4.2.b" and substitute "Gage" in the Inspection Method column.

* Add Minor: "206 Marking missing, incomplete or illegible 0.65% 3.2 Visual"

Add Note: "(a) See 4.4.3.1"

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* 4.4.2.5: Add Major: "109 Salt spray test a/ 3.2 Visual"

Add Note: "(a) See 4.4.3.1"

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* 4.4.2.6: Delete Minor 201 in its entirety.

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* 4.4.3: At the end of the sentence, add: "and the following:"

Add new paragraph 4.4.3.1 as follows:

"4.4.3.1 Salt spray test. (See 4.5.6) For this test, a lot of anodized parts shall be the quantity anodized during 8 hours (max) of production. From each lot of anodized fins or housings, three units shall be randomly selected for the test.

If the fins and housings are part of the same anodizing lot, three units of each type shall be tested (see 6.10). In the event that only one type of component fails the test, only the units of the same type in that lot shall be rejected.

After 10 consecutive production lots pass the test, sampling may be reduced to twice weekly for 2 weeks, and if successful, further reduction to one test every 2 weeks is authorized. If a failure occurs during any phase of reduced sampling, then the sampling shall revert back to the original plan, until 10 consecutive lots pass, etc."

4.5.3.1: Delete in its entirety.

4.5.3.1.1: Delete in its entirety.

4.5.3.2: Delete in its entirety and substitute the following:

"4.5.3.2 Tension test. Following final heat treatment, one specimen shall be submitted per 1,000 pound of stock from both housing stock and fin stock. The fin specimens shall be the largest standard flat-type specimens obtainable from the blade area and the housing specimens shall be the largest standard round-type specimens obtainable from housing stock. Specimens shall be taken longitudinally. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the applicable solution heat treatment batch or press-quench lot. The test criteria for all tensile specimens shall be as prescribed by ASTM Method E-8. The number of specimens and their location is to verify compliance with the mechanical properties requirements. The Government shall not be restricted from testing additional specimens at any location to determine that the mechanical properties meet the drawing requirements for yield strength and elongation."

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4.5.3.2.1: Delete paragraph in its entirety.

4.5.3.2.2: Delete paragraph in its entirety.

4.5.3.3: Delete paragraph in its entirety and substitute the following:

"4.5.3.3 Microstructure examination.

4.5.3.3.1 Fin stock. From the fin blade adjacent to each of the flat samples submitted for tension testing, one wafer approximately 1/2 inch square shall be prepared for metallographic examination from undeformed material. When suitably prepared, the specimens shall be etched with Keller's reagent and examined by scanning under a metallurgical microscope at a minimum magnification of 500 diameters. Overheating shall be evidenced by the presence or rosettes in the microstructure or incipient melting at the grain boundaries (see 6.5). Evidence of the above shall cause rejection of the applicable solution heat threat batch or press-quench lot.

4.5.3.3.2 Housing stock. Metallographic specimens from housing material shall be cut from a transverse section at either end of each round specimen submitted for tension testing. Preparation and examination for overheating, porosity, etc. shall be as described in 4.5.3.3.1."

4.5.3.4: Delete in its entirety.

4.5.4 Hydrostatic test.

Line 10: Delete the following sentence: "The section where the fin is assembled to the housing shall be fully supported on the outside diameter during test."

Line 24: Delete the following sentence: "For first article inspection only, the test pressure shall be increased until the housing ruptures and the maximum pressure shall be recorded for information purposes."

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4.5.5.2: Delete paragraph in its entirety and substitute the following:

"4.5.5.2 Excessive flash holes. The sample fin housings shall be visually inspected for the presence of excess flash holes."

Add new paragraph 4.5.6 as follows:

"4.5.6 Blade security test: The housing portion shall be held securely in a fixture while 375 ± 25 inch-pounds torque is applied for 3 seconds to the blade portion of the assembly. Rotation of the blades around any housing shall be cause for rejection. If the blade cants are contacted during testing, the assembly must be discarded whether or not the assembly passes."

* Add new paragraph 4.5.7 as follows:

"4.5.7 Salt spray test. (See 4.4.3.1) Unless otherwise specified, equipment and procedures for testing shall conform to ASTM B-117. Component parts (not test panels) shall be used. Before testing, the parts should be washed in distilled or deionized water and dried with a soft, clean cloth. The test specimens shall be fixtured between 6 and 30 degrees from vertical during testing. The test duration and concentration of the salt spray shall be as specified on drawings 11726795 or 11726889 as applicable.

After testing, the parts shall be rinsed in clean water and dried by air. The significant surface shall be examined to determine that th ere are no corrosion spots larger than 0.010 inch in diameter, and that there are no patchy dark gray areas of spots, streaks, or marks. The significant surface shall be the entire area within approximately 45 degrees of the portion of the component which faced straight up in the chamber, excluding any area within 0.062 inch of stamp marks, edges, and electrode contact marks. Failure of a sample for any of the reasons shall cause rejection of the entire applicable quantity."

6.3, Line 4, Delete: "DRDAR-QAR-I" and substitute "AMSTA-AR-QAA-C."

6.8, Add the following: "Direct submissions to Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSTA-AR-QAA-C, building 62, Picatinny Arsenal, NJ 07806-5000."

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* Add new paragraph 6.10 as follows:

"6.10 Anodizing process. The supplier should be thoroughly familiar with all relevant aspects of MIL-A-8625 (AR); e.g., the impact of the anodic coating on thread dimensions (which must be checked after anodizing).

Ferrous materials (e.g., steel wire wheels) should not be used for deburring.

Anodizing fins and housings (or other items) simultaneously (mixed loads) is not recommended due to the possible need to compromise process parameters.

Although test panels are not permitted for salt spray testing, they are acceptable for maintaining control and verifying compliance with other requirement of the process, such as unit coating weight, coating thickness, etc.

During the anodizing process, it would serve the customer's long term needs best if components were fixtured in a way that would minimize or completely prevent the outside surfaces from making any contact with the electrodes/racks, while at the same time permitting easy fluid flow and drainage in the internal area. It is the external surfaces which need the most continuous coating possible."

The margins of this amendment are marked with an asterisk, or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last pervious amendment.

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